Blood Sugar & Diabetes: Reading

Biology A · Unit 6 · Level 3 (closer to the class work) · goes with Unit 3 Diabetes: A1 CDC data, A3/A6 Homeostasis, ClEvR quizzes, A7 Risks, A8 Solutions, A10 Review, Diabetes Test

Name: _____

Keeping Blood Sugar in Balance

Homeostasis is the body keeping its inside conditions balanced, the way a thermostat keeps a room at the right temperature. Blood sugar (blood glucose) goes up after we eat. A feedback loop keeps it in a healthy range.

When blood sugar is high, the pancreas releases insulin. Insulin works like a key: it unlocks cells so glucose can go in, and blood sugar goes down. When blood sugar is low, the pancreas releases glucagon, which lets stored sugar out into the blood, and blood sugar goes up.

Some foods raise blood sugar fast. Sugary drinks and white bread make it spike. Beans, vegetables and whole grains raise it slowly because of fiber. Sugar in whole fruit comes with fiber too.

Diabetes is a disease where blood sugar stays too high for too long. In type 1 diabetes, the pancreas makes little or no insulin; it often starts young and is not caused by food choices. In type 2 diabetes (90-95% of cases), cells stop answering insulin well. In ClEvR #2, diabetic blood sugar three hours after eating was 200 or more, compared with about 120-140 without diabetes.

Risk factors can be genetic (family history, which we cannot change) or environmental (food, activity, sleep, money and where people live, some of which can change). CDC data shows higher rates where people have less money or schooling. Solutions can help people and whole communities: drinking water instead of sugary drinks, more chances to move, clear food labels and medical treatment.

Check your reading

1. What does insulin do?

2. How is type 1 different from type 2 diabetes?

3. Give one genetic and one environmental risk factor.